2 ALARM NUMBERS AND BRIEF DESCRIPTIONS

Number	Alarm type	LED type (Classification by Red LED)
000 to 299	Program or setting alarm	□□□□ ■□□ or □□□□ □□■
300 to 399	Pulse coder alarm	□□□□ ■■□
400 to 499	Servo alarm	□□□□ ■□■ or □□□□ □■■
500 to 599	Overtravel alarm	□□□□ ■□□ Same as PS alarm
-	System alarm or I/O Link alarm	□□□□ □■□ or □□□□ ■■■

Program or setting alarms (PS alarms)

No.	LED display	Description	Countermeasure
000	□□□■ ■□□	A parameter that requires power-down has been specified.	Turn the power off, then back on.
011	□□■□ ■□□	The specified feedrate is zero.	Check the feedrate parameter specified with a function code.
013	□□■□ ■□□	The specified feedrate (maximum feedrate) is zero.	Check the value of parameter No.043, which indicates the maximum feedrate that can be specified.
070	□□■■ ■□□	More than 32 blocks have been registered for a buffering operation.	Reduce the number of registered blocks to 32.
090	□■□■ ■□□	Reference position setting cannot be executed normally.	Move the tool in the direction of reference position return in jog mode at a speed that causes the servo position error to exceed 128. Then, specify another reference position setting.
093	□■■□ ■□□	(1) A first to third reference position return cannot be executed because the reference position has not yet been established. (2) The reference position external setting was executed without Absolute pulse coder.	(1) Set the reference position. (2) Absolute pulse coder must be used.

No.	LED display	Description	Countermeasure
224	□ ■ ■ □ ■ □ □	The reference position has not yet been established. This occurs only when the ZRTN bit of parameter No.001 is set to 0.	Set the reference position.
250	□ ■ ■ ■ ■ □ □	Input data 1 or command is invalid.	Check input data 1, specified with a function code.
251	□ ■ ■ ■ ■ □ □	Input data 2 is invalid.	Check input data 2, specified with a function code.
254	□ ■ ■ ■ ■ □ □	A function code or mode is invalid.	Check the command code, specified with a function code. Check the mode.
255	■ □ □ □ ■ □ □	Operation cannot be activated because an invalid mode is specified or because block execution is in progress.	Check the mode. Check whether a block is being executed.
290	■ □ □ ■ ■ □ □	The interface switch signal (DRC) was switched during block execution.	Switch the signal after block execution stops.
291	■ □ ■ □ ■ □ □	The speed of an axial movement specified by an external pulse has exceeded the upper limit. This occurs only when the EPEXA bit of parameter No.001 is set to 1.	Check the speed specified by the external pulse. Check the magnification of the external pulse (parameters No.062 and 063).
292	■ □ ■ ■ ■ □ □	A checksum error for the nonvolatile memory was detected.	Parameters are cleared. Set the parameters again. If this alarm subsequently recurs, replace the unit.
293	■ ■ □ □ ■ □ □	Software version between FROM on CPU and EPROM is same.	Remove the EPROM.
294	■ ■ □ ■ ■ □ □	CRC check alarm of EPROM was detected.	Remove the EPROM.

Pulse coder alarms

No.	LED display	Description	Countermeasure
300	□□□■ ■■□	A communication error (DTER) for the serial pulse coder was detected.	Check the continuity of the signal cable. If the cable is normal, the pulse coder or servo amplifier module may be defective. This error can also be caused by external noise. See the chapter on noise reduction in "Descriptions."
301	□□□■ ■■□	A communication error (CRCER) for the serial pulse coder was detected.	Check the continuity of the signal cable. If the cable is normal, the pulse coder or servo amplifier module may be defective. This error can also be caused by external noise. See the chapter on noise reduction in "Descriptions."
302	□□□■ ■■□	A communication error (STBER) for the serial pulse coder was detected.	Check the continuity of the signal cable. If the cable is normal, the pulse coder or servo amplifier module may be defective. This error can also be caused by external noise. See the chapter on noise reduction in "Descriptions."
303	□□■□ ■■□	An LED disconnection (LDAL) was detected in the serial pulse coder.	Turn the power off. If this alarm recurs when the power is reapplied, replace the motor.
304	□□■□ ■■□	A mispulse alarm (PMAL) for the serial pulse coder was detected.	Turn the power off. If this alarm recurs when the power is reapplied, replace the motor.
305	□□■□ ■■□	A miscount alarm (CMAL) for the serial pulse coder was detected.	Turn the power off. If the alarm recurs when the power is re-applied, replace the motor. Even if the alarm does not recur, restart the operation from reference position return.
306	□□■■ ■■□	The motor has overheated (OHAL).	This alarm is issued when the motor has overheated, causing the thermostat to trip. Possible causes include an excessively high ambient temperature and excessively strict operating conditions. Check the actual cause. If it occurs again when the motor is cooled, the motor or servo amplifier may have failed. Replace the faulty motor or servo amplifier.
308	□□■□ ■■□	A soft phase alarm (SPHAL) was detected.	Turn the power off. This alarm may be caused by noise.

No.	LED display	Description	Countermeasure
319	□ ■ □ □ ■ ■ □	When the absolute pulse coder is used, the motor has not yet rotated through more than one turn after the first power-up.	Cause the motor to rotate through more than one turn in jog feed mode, then turn the power off then back on.
350	□ ■ □ ■ ■ ■ □	The battery voltage of the absolute pulse coder is low.	Replace the battery. Restart the operation from reference position return.
351	□ ■ ■ □ ■ ■ □	The battery voltage of the absolute pulse coder is low. (warning)	Replace the battery.

Servo alarms

No.	LED display	Description	Countermeasure
400	□□□■ ■□■	The servo motor has overheated. (estimated value)	The motor operation condition may be too severe. Check the operation condition.
401	□□■□ ■□■	DRDY(Drive Ready signal) becomes 0. The servo amplifier is not ready.	
404	□■□□ ■□■	The regenerative discharge unit has overheated.	This alarm is issued when the average regenerative discharge energy is too high (when the acceleration/deceleration frequency is too high, for example).
		(1) When the separate regenerative discharge resistor is not used. For the SVM1-4*i* and SVM1-20*i*, check whether the CXA20 connector is short-circuited by using a dummy connector. For the SVM1-40*i* and SVM1-80*i*, check whether the CXA20 and CZ6 connectors are short-circuited by using dummy connectors. (2) The average regenerative discharge energy may be too high. Decrease the acceleration/deceleration frequency. (3) The separate regenerative discharge unit may not be connected properly. Check the connection. (4) The thermostat of the separate regenerative discharge unit may be defective. Disconnect the separate regenerative discharge unit, then check the thermostat. If the thermostat is open even through the separate regenerative discharge unit is cool, replace the separate regenerative discharge unit. (5) The resistor of the separate regenerative discharge unit may be defective. Disconnect the separate regenerative discharge unit, then check the resistance. If it does not fall in the range of the predetermined resistance "20%, replace the separate regenerative discharge unit. (6) If from (1) to (5) are not the cause of the alarm, replace the servo amplifier module.	
405	□■□■ ■□■	Reference position return could not be executed correctly.	Re-execute reference position return.
		If a value in the range of 4 to 96 is set for parameter No. 032 (CMR), an alarm may be issued. In this case, prevent an alarm from being issued by setting N405(bit4 of parameter No. 001) to "1".	
409	□■■□ ■□■	Unexpected disturbance torque has been detected.	Look for the mechanical cause of the abnormal load. If no mechanical cause is found, specify a larger value for the parameter.

No.	LED display	Description	Countermeasure
410	□ ■ ■ ■ ■ □ ■	The servo position error in the stop state is larger than the value specified in parameter No.110.	Determine the mechanical cause of the large position error. If no mechanical cause is found, specify a larger value for the parameter.
411	□ ■ ■ ■ ■ □ ■	The servo position error during movement is larger than the value specified in parameter No.182.	Determine the mechanical cause of the large position error. If no mechanical cause is found, apply any of the following countermeasures: • Specify a larger value for the parameter. • Specify a lower feed rate. • Increase the time constants.
412	■ □ □ □ ■ □ ■	Over Current Alarm has been issued.	This alarm is issued when an excessively large current flows in the main circuit.
		(1) Check whether the motor ID number is correctly set in parameter No.125. (2) Check whether the standard values (see Appendix B) are specified in the current control parameters for servo control. Correct current control is possible only when the standard values are specified for the following parameters: No.70, 71, 72, 78, 79, 84, 85, 86, 87, 88, 89, 90, 99, 118, 119 (3) Disconnect the power line from the servo amplifier module connector. - Then, release the emergency stop state. If the over-current alarm continues to be issued, replace the servo amplifier module. - If no over-current alarm is issued, go to (4). (4) Check the insulation between the ground and each of U, V, and W. - If the insulation is satisfactory, go to (5). If a short-circuit is detected, disconnect the power line from the motor connector. Then, check the insulation between the ground and each of U, V, and W of the motor. - If a short-circuit is found between the ground and U, V, or W of the motor, replace the motor. - If the insulation is satisfactory, replace the power line. (5) Connect the power line. Observe the waveform of the motor current (IR, IS) while the motor is accelerating or decelerating. See the description of the checking of the motor current in Appendix E, "Servo Check Board". - If the motor current (IR, IS) does not exhibit a normal sine wave, replace the servo amplifier module. (6) Check if the motor current (IR, IS) waveforms include noise. - When noise is included, take an action such as making a connection to shield ground. - When noise is not included, replace the servo amplifier module. If (1) to (6) above are not the cause of the alarm, the pulse coder, command cable or internal hardware of the CNC may be defective.	

No.	LED display	Description	Countermeasure
413	■□□■ ■□■	DC Link Over Voltage Alarm has been issued.	This alarm is issued when the DC voltage of the main circuit power is too high.
		(1) When SVM1-4*i* or SVM1-20*i* is used, and a separate regenerative discharge unit is not used, check the specification to see if regenerative energy per one time does not exceed the allowable regenerative energy of the servo amplifier module. (2) For the SVM1-40*i* and the SVM1-80*i*, when the separate regenerative discharge resistor is not used, check whether the CZ6 connector is short-circuited with a dummy connector. (3) The supply voltage for dynamic power may exceed the rated value. Check the voltage. If the voltage is too high, reduce the voltage to an appropriate level. (4) The regenerative discharge unit may not be properly connected. Check the connection. (5) The resistance of the separate regenerative discharge unit may be abnormal. Disconnect the separate regenerative discharge unit, then check the resistance. If the resistance is not within ±20% of the predetermined resistance, replace the separate regenerative discharge unit. If from (1) to (5) are not the cause of the alarm, replace the servo amplifier unit.	
414	■□■□ ■□■	DC Link Low Voltage Alarm has been issued.	This alarm is issued when the DC voltage of the main circuit power is too low.
		(1) 190 ms or longer may pass from the time when both the *ESP of the built-in DI and the *ESP of the I/O link interface signal are canceled until the external magnetic contactor inserted into the input for motive power turns on(including the operating time of the magnetic contactor). The magnetic contactor must turn on within 100 ms. (2) The external circuit breaker may be turned off. Check the circuit breaker. (3) The supply voltage for dynamic power is lower than the rated value. Check the voltage. If the voltage is too low, increase it to an appropriate level. (4) The external magnetic contactor may not be connected properly. Check the connection. If from (1) to (4) are not the cause of the alarm, replace the servo amplifier unit.	

No.	LED display	Description	Countermeasure
416	■□■■ ■□■	Inverter IPM Alarm has been issued.	(1) Check whether the cooling fan is not stopped. (2) Check whether the motor is used within related current. (3) Check whether the temperature in locker is higher. Check the fin or filter in locker. (4) Check whether the temperature is higher around the servo amplifier module. (5) Disconnect the power line from the servo amplifier module connector. Then, release the emergency stop state. - If the IPM alarm continues to be issued, replace the servo amplifier module. - If no IPM alarm is issued, go to (6). (6) Disconnect the power line from the servo amplifier module connector. Then, check the insulation between the ground and each of U, V, and W. - If the insulation is satisfactory replace the servo amplifier module. - If the insulation is not satisfactory, go to (7). (7) Disconnect the power line from the servo motor. Then check the insulation. - If the insulation of servo motor is not satisfactory, replace the servo motor - If the insulation of power line is not satisfactory, replace the servo amplifier module.
417	□□■■ □■■	A parameter has been specified incorrectly.	Check the following parameters: - When the both No.030 and No.125 are set for motor ID number, this alarm is issued.it is necessary to set motor ID number on No.125 and to set 0 on No.030. - No.031: the parameter for direction of rotate is not selected in “111” or “–111”. - No.106: Is the denominator of the number of pulses per single revolution of the motor 0? - No.107: Is this parameter too large? - No.179: Is this parameter out of range (8388607 lower or equal)? - Is No.107 (position gain)/No.105 or No.179 (number of pulses per single revolution of the motor) less than 0.488? - No.180: Is the specified reference counter capacity 0 or a negative value?

No.	LED display	Description	Countermeasure
423	■■□■ ■□■	The specified speed exceeds 32767000 detection units per second.	Re-examine the CMR and speed settings.
425	■■■□ ■□■	The fan for internal agitation has stopped. This alarm is issued when the fan motor built into the servo amplifier module has failed. The fan motor is consumable. For the explanation about its replacement procedure, see Part IV "Maintenance of Servo Amplifier Unit." (1) Check that the fan is not clogged with foreign matter. (2) Check that the power connector of the fan is connected properly. (3) Replace the fan or servo amplifier module.	
446	■■■■ ■□■	The external pulse input line is disconnected.	Connect the external pulse input signal correctly.
447	□■■■ ■□■	The velocity deviation is too high (velocity control)	Check the actual velocity. See the settings of parameter No.136.
449	□□□■ □■■	No.080 is not corrected. (This parameter is out of range from 0 to 8010)	Set No.080 correctly (from 0 to 8010).
601	□□■□ □■■	The fan for cooling the external radiator fin has stopped. This alarm is issued when the fan motor for inverter radiator has failed. The fan motor is consumable. For an explanation of the replacement procedure, see Part IV, "Maintenance of Servo Amplifier Unit." (1) Check that the fan is not clogged with foreign matter. (2) Check that the power connector of the fan is connected properly. (3) Replace the fan or servo amplifier module.	

Overtravel alarms

No.	LED display	Description	Countermeasure
500	■■■■ ■□□	The positive stroke limit has been exceeded.	Check whether *+OT and *-OT are connected correctly. Check whether a correct move command is specified. Move the tool in the opposite direction in jog mode, then perform a reset.
501	■■■■ ■□□	The negative stroke limit has been exceeded.	
510	■■■■ ■□□	The positive soft stroke limit has been exceeded.	Check whether appropriate values have been specified for parameters No.142 and 143. Check whether a valid move command is specified. Move the tool in the opposite direction in jog mode, then perform a reset.
511	■■■■ ■□□	The negative soft stroke limit has been exceeded.	